Problem 2: Finding Complexity using Counter method

Started on	Monday, 24 August 2026, 12:36 PM
State	Finished
Completed on	Monday, 24 August 2026, 6:39 PM
Time taken	6 hours 2 mins
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

void func(int n)
{
    int count = 0;

    if (n == 1)
    {
        count++;
    }
    else
    {
        count++;  // i = 1

        for (int i = 1; i <= n; i++)
        {
            count++;  // i <= n

            count++;  // j = 1

            for (int j = 1; j <= n; j++)
            {
                count++;  // j <= n

                count++;  // break

                break;
            }

            count++;  // i++
        }

        count++;  // final failed i <= n
    }

    printf("%d", count);
}

int main()
{
    int n;
    scanf("%d", &n);

    func(n);

    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

Back to Course